

Where SatQuery be BETTER than existing models

1. Query → Complete Analysis, not just Query → Answer

Most RS-VLMs are primarily:

Image + Question → Answer

SatQuery should become:

Question → Understand → Plan → Select tools → Analyze → Verify → Map → Explain

For example:

"Find newly constructed buildings within 500 m of the flooded area between June and August."

SatQuery shouldn't just describe the image.

It should determine:

Temporal analysis

+

Flood detection

+

Building detection

+

Grounding

+

Spatial buffer

+

Intersection

+

Evidence

That's a **system-level capability**, and it's directly aligned with the agentic requirement in the PS.

 **This should be our #1 advantage.**

2. **Dynamic Specialist Selection**

Instead of forcing one giant model to solve everything:

User



SatQuery Controller



"What is the best way to solve this?"

Then:

VQA question

→ RS-VLM

"Find all ships"

→ open-vocabulary detector

"Outline the lake"

→ segmentation model

"What changed?"

→ temporal/change model

"Compare optical + SAR"

→ multimodal specialist

"Calculate vegetation health"

→ spectral/GIS engine

This is much more sensible than expecting one model to be simultaneously excellent at VQA, segmentation, change detection, SAR interpretation and GIS.

3. **Optical + SAR Reasoning**

This is one of our biggest opportunities.

SkySense already demonstrates sophisticated multimodal and spatiotemporal EO representation learning, including optical + SAR and temporal sequences.

EarthGPT also expanded multimodal input beyond RGB toward SAR/infrared, while the RS-VLM literature increasingly addresses multimodal perception.

So we **cannot claim**:

"Nobody combines optical and SAR."

That's false.

But we can aim for something more useful:

Existing model:

"Here are representations from optical + SAR."

SatQuery:

"Given this user's question, determine whether optical, SAR, or both are necessary, perform the appropriate analysis, compare the evidence, and explain what each sensor contributes."

For example:

Optical image: heavy cloud cover.

SatQuery could recognize:

Optical confidence ↓

SAR reliability ↑

and route accordingly.

That's a much more **decision-oriented** use of multimodal data.

4. Sensor-Aware Reasoning

This is an even deeper version of #3.

Instead of simply:

"I have two images."

SatQuery should understand:

"What information does each sensor provide?"

For example:

Optical

→ color

→ spectral information

→ visual context

SAR

→ structural/backscatter information

→ works through clouds

→ day/night capability

Then the system can reason:

"The optical image is cloud-obscured, so SAR provides stronger evidence for this particular question."

That's not merely multimodal input.

It's **sensor-aware decision making**.

5. Temporal Reasoning Beyond Pixel Difference

Existing change-detection models are already very good at detecting changes.

So we shouldn't compete merely on:

"We can detect change."

Instead:

Level 1

Where did something change?

Level 2

What changed?

Level 3

Which objects changed?

Level 4

How much changed?

Level 5

What does the change mean?

Level 6

What should the user investigate next?

For example:

"What changed in this area?"

SatQuery could produce:

2024

↓

Agricultural land

2026

↓

Built-up structures

Change:

+18 buildings

Affected area:

4.7 hectares

Evidence:

Highlighted regions

Confidence:

0.91

That is much more valuable than a raw change mask.

And temporal RS-VLM research is already moving toward language-based temporal understanding, so again, the novelty must be in the **whole workflow**, not simply "we do temporal reasoning."

6. Better Spatial Grounding

This could become a serious differentiator.

Some models are already strong here.

For example, recent RSThinker results reported extremely strong visual grounding numbers on VRSBench and DIOR-RSVG.

So:

 Don't claim:

"SatQuery invented satellite grounding."

Instead:

SatQuery advantage:

Grounding + GIS semantics + natural language + evidence.

User:

"Show the buildings within 300 metres of this river."

System:

River



Segment



Buildings



Coordinates



300m buffer



Spatial intersection



GeoJSON



Map

That's significantly more useful than simply returning a bounding box.

7. 🌍 From AI Answer → Actual GIS Result

This is one area where I think SatQuery can become **extremely practical**.

A VLM might say:

"There appear to be several water bodies."

SatQuery should be able to return:

Water bodies detected: 7

Total area: 12.4 km²

Largest:

4.8 km²

GeoJSON:



Map layer:



Evidence:



And potentially allow:

Download result.

So we are transforming:

AI interpretation

into

usable geospatial intelligence.

8. Automated Spectral Reasoning

This could be another strong practical advantage.

Instead of making users know:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

they say:

"Show areas with poor vegetation health."

SatQuery can determine:

Intent



Vegetation health



Relevant spectral bands



NDVI calculation



Threshold/analysis



Map



Explanation

The important innovation isn't inventing NDVI.

Obviously we don't.

The innovation is:

Natural language → appropriate geospatial analytical operation.

9. Multi-step Queries

This could become one of our signature capabilities.

Simple query:

"Find buildings."

Complex query:

"Find new buildings constructed after 2024 within 1 km of roads in areas where vegetation decreased significantly."

Now the system has to perform:

Temporal change

+

Building detection

+

Road detection

+

Distance calculation

+

Vegetation index

+

Thresholding

+

Spatial filtering

+

Result integration

This is where a **query-driven agentic architecture** can become much more powerful than a single VLM.

10. Evidence Before Confidence

This is one of my favorite potential SatQuery features.

Most AI systems essentially say:

"I'm 92% confident."

But why?

SatQuery should ideally show:

ANSWER

New construction detected.

EVIDENCE



Highlighted buildings

TEMPORAL EVIDENCE

2024 → absent

2026 → present

SENSOR EVIDENCE

Optical ✓

SAR ✓

CONFIDENCE

91%

REASON

Detected in both temporal and cross-sensor evidence.

This makes the system much more defensible for an ISRO-type evaluation.

The PS explicitly expects **visual evidence and confidence information**.

11. Honest Uncertainty

This could be more important than squeezing another 1–2% benchmark accuracy.

Imagine:

"Is there an oil spill?"

The model isn't sure.

Bad SatQuery:

"Yes, there is an oil spill."

Better:

"Possible oil-like anomaly detected, but confidence is low because the optical image is cloud-obscured and the SAR evidence is ambiguous."

Then:

"Recommend reviewing the highlighted region with additional temporal imagery."

That is much closer to a useful scientific assistant.

And the RS literature continues to identify hallucination and domain mismatch as major problems for VLMs.

12. Self-Verification

This could be one of our strongest architectural ideas.

Instead of:

Model → Answer

use:

Model



Answer



Verifier



Evidence check



Spatial consistency check



Confidence



Final answer

For example:

"There are 18 buildings."

Verifier:

Detection count = 18
Grounding count = 18
Visual evidence = valid

→ Accept.

If:

Detection = 18
Grounding = 13

→ Flag uncertainty / rerun.

That turns SatQuery from:

generator

into:

generator + verifier.

13. 🧠 Model-Agnostic Architecture

This is a strategic advantage.

Don't hard-code:

SatQuery = GeoChat

Instead:

SatQuery
|
├── Model A today
├── Model B tomorrow
├── Model C later
└── custom model eventually

The router chooses based on capability.

This means if a better RS-VLM appears next year:

replace specialist → keep SatQuery architecture.

That is much more future-proof.

14. 🏆 What I DON'T Think SatQuery Should Try to Beat

This is equally important.

We should **not** try to beat every SOTA model at its own specialty.

For example:

SkySense

Already extremely strong as a multimodal/spatiotemporal RS foundation model.

SkySense-O

Strong open-world RS interpretation and reported improvements over several RS models.

RSThinker

Very strong grounding/reasoning results in its reported benchmarks.

UniRS

Strong unified multi-temporal VLM capabilities and competitive VQA results.

So our strategy should be:

Use the best available specialist rather than trying to recreate it.

🧠 The real SatQuery advantage

Here's the key distinction:

Existing approach

SatQuery goal

Image → answer	Query → analysis → answer
One model	Multiple specialists
Fixed task	Dynamic task selection
Mostly visual answer	Spatially grounded result
One modality	Sensor-aware multimodal reasoning
Raw prediction	Evidence + confidence
Detection	Detection + GIS operation
Change mask	Change → explanation → impact
Model-centric	User-query-centric
AI output	Actionable geospatial intelligence

And this fits the actual SIH problem statement unusually well.

If I had to choose the TOP 7 SatQuery advantages

I would prioritize these:

1. Intelligent Query-to-Workflow

Natural language → automatically determine what needs to happen.

2. Dynamic Specialist Routing

Choose the right model/tool rather than forcing one model to do everything.

3. Temporal + Cross-Sensor Reasoning

Especially bi-temporal + optical/SAR.

4. Grounded Geospatial Answers

Not just text—boxes, masks, coordinates, GeoJSON, maps.

5. Multi-Step Geospatial Reasoning

One natural-language query can trigger multiple analytical operations.

6. Evidence + Verification + Uncertainty

Don't just say *what*; show *why* and how reliable it is.

7. Actionable GIS Intelligence

Turn satellite observations into measurements, spatial relationships, reports and downloadable layers.



And this is the BIG idea

I wouldn't position SatQuery as:

“A better VLM than GeoChat.”

or:

“A better model than SkySense.”

That is the wrong battlefield.

I'd position it as:

“A system that makes the capabilities of many advanced EO models usable through one intelligent natural-language interface.”

Think of it like this:



